

NAME: _____ MATRIC NUMBER: _____

SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT 2 SEMESTER 1 2022-2023

MS1016 – THERMODYNAMICS OF MATERIAL

8 November 2022

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
 2. Answer **ALL** questions.
 3. This is an open book test.
 4. All questions have **ONLY ONE** correct answer for the MCQ questions.
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PART I – SHORT QUESTIONS

- (1) Explain the Gibbs Phase rule. Also determine if 4 phases can exist in equilibrium in a binary system?

(20 marks)

Gibbs Phase Rule :

$$f = 2 - P + C$$

where C is the number of components and P, the number of phases. f is the degree of freedom (number of parameters that can be fixed) to solve the chemical potential equilibrium equations and determine the composition of the phases in equilibrium.

Eg. For 2 components and 2 phases, $f = 2$. By fixing the temperature and pressure, the chemical compositions of the 2 phases can be determined.

For a binary system with 2 components, and 4 phases, $f = 0$. This means that no parameter can be fixed. Hence 4 phases cannot exist together in equilibrium.

2. Described the significance of the Raoult's and Henry's Law and explain how these laws are used for in solid solution.

(20 marks)

The activity coefficient, γ is defined as the ration of the activity over mole fraction,

$$\gamma = \frac{a}{X}$$

where a is the activity and X , the mole fraction. The **significance** of Raoult's and Henry's law is that calculation of the chemical potential of a component in a dilute solid solution can be simplified to the involvement of only the mole fraction. Given the chemical potential of a component, μ is

$$\mu = \underline{G} + RT \ln a$$

Raoult's law states that the activity of the major component in a dilute solution is approximately ideal and may be approximated to the mole fraction. Hence the chemical potential may be written as:

$$\mu = \underline{G} + RT \ln X$$

For the minor component in the dilute solution, the activity equals to γX . Henry's law states that γ may be approximated to be a constant because the composition is X is low. Hence

$$\mu = \underline{G} + RT \ln \gamma X$$

Therefore. Using Raoult's law and Henry's law, the chemical potentials of the components in a dilute solution may be calculated without using the activity.

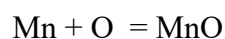
3. The element manganese, Mn can be extracted from its oxide, MnO. Suggest a suitable material to reduce MnO to Mn. Determine the appropriate temperature and equilibrium partial pressure of oxygen for this process.

(20 marks)

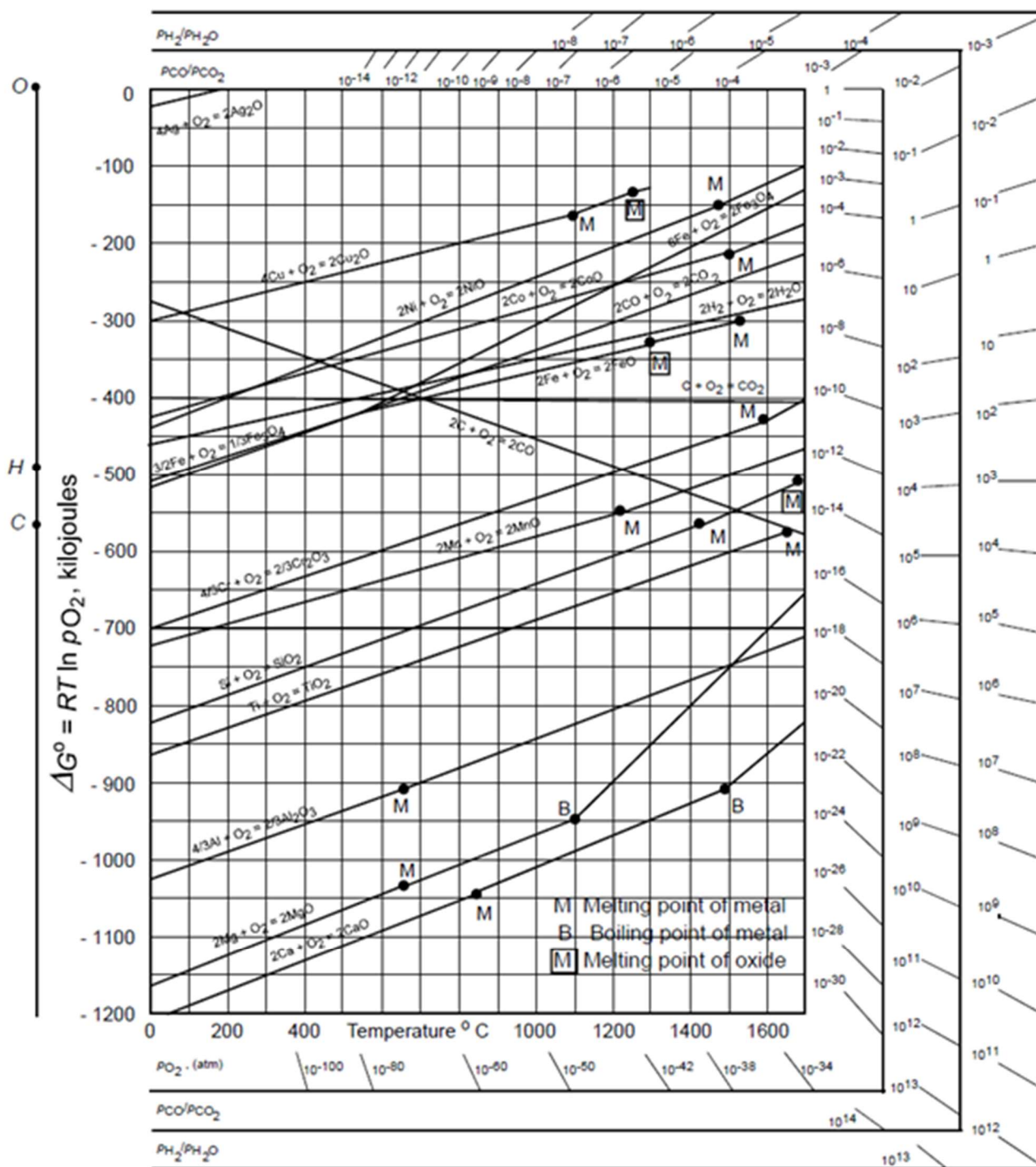
At any temperature, the oxide with the lower standard Gibbs free energy of oxidation is more stable. Therefore by inspecting the Ellingham diagram, any of the oxides with the standard Gibbs free energy lower than MnO can be used to extract Mn of MnO.

Carbon in the form of CO is most suitable and as a gas, it can interact readily with MnO during the reduction process. Inspecting the Ellingham diagram, the minimum temperature for the reduction of MnO is 1400°C.

From the Ellingham diagram, the equilibrium pressure for the oxidation of MnO,



is approximately 10^{-16} atmosphere.



PART II MCQ QUESTIONS

- (4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(40 marks)

- (a) Henry's Law states that

- A the activity coefficient of the solvent in a dilute solution is a constant
- B the activity coefficient of the solute in a dilute solution is a constant
- C the activity coefficient of a component in a concentrated solution is a constant
- D the activity of the solute in a dilute solution equal to the mass fraction
- E the activity of the solute in a dilute solution equal to unity

ANS: B

- (b) The Clausius-Clapeyron equation is given as:

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{\underline{S}^{\beta} - \underline{S}^{\alpha}}{\underline{V}^{\beta} - \underline{V}^{\alpha}} = \frac{\underline{\Delta S}}{\underline{\Delta V}}$$

The melting point of a certain solid was found to decrease when the external pressure is increased. Which of the following deduction is probably correct?

- A dP/dT is negative
- B it is more difficult to melt the material under pressure
- C the volume of the material expands upon melting
- D dP/dT is positive
- E None of the above

ANS: A

- (c) In a reaction, the ratio of the activities of the products over the reactants at equilibrium is called the:

- A standard molar Gibbs free energy
- B equilibrium partial pressure
- C equilibrium constant
- D Gibbs free energy of mixing
- E activity coefficient

ANS: C

(d) The driving force for dilute solutions to be formed in a binary alloy where $\Delta H_{\text{mix}} > 0$ is to increase the

- A Gibbs free energy
- B constant pressure heat capacity
- C internal energy
- D entropy
- E enthalpy

ANS: D

(e) Which of the following is NOT a single component system?

- A water and ice
- B white and gray tin
- C graphite and diamond
- D amorphous Si glass and crystalline Si glass
- E Carbon monoxide (CO) and Carbon Dioxide (CO₂)

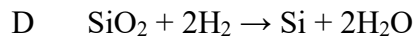
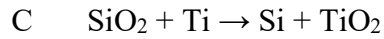
ANS: E

(f) The activity of a component replaces the mole fraction in a binary solid solution because of the _____ solution.

- A High Temperature and Low Pressure
- B High Temperature and High Pressure
- C Low Temperature and High Pressure
- D Low Temperature and Low Pressure
- E none of the above

ANS: C

(g) From the Ellingham diagram, which of the following reduction reaction is likely used for the extraction of Si from SiO_2 ?



ANS: E

(h) Refer to the Ellingham diagram. Estimate the O_2 pressure for the oxidation of Mn at $1000^\circ C$?

A 10^{-21} atmospheric pressure

B 10^{-22} atmospheric pressure

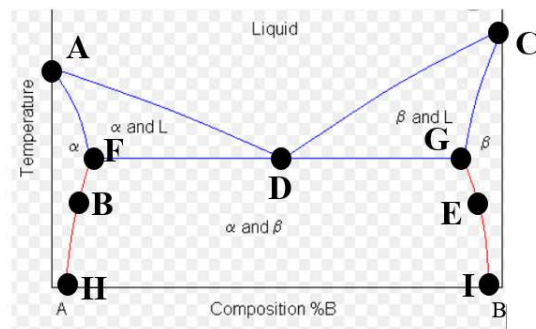
C 10^{-23} atmospheric pressure

D 10^{-24} atmospheric pressure

E 10^{-25} atmospheric pressure

ANS: D

(i) Figure below is the binary equilibrium phase diagram consisting of elements A and B. List the points where 3 phases are in equilibrium.



A A, C, H, and I only

B F, D and G only

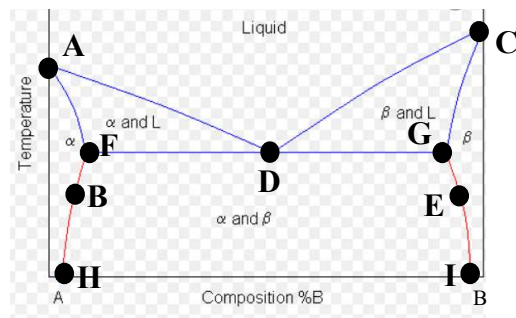
C B and E only

D A, C, F and G only

E A, C, F, D and G only

ANS: B

- (j) Figure below is a binary phase diagram of components A and B. The solubility limit of A in B labelled in the phase diagram is represented by the letter:



ANS: E